A SYNOPSIS ON

SYSTEM FAILURE ANALYZER

Submitted in partial fulfilment of the requirement for the award of the degree of

BACHELOR OF COMPUTER APPLICATION

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SEPTEMBER-2025

CANDIDATE'S DECLARATION

I/We hereby certify that the work which is being presented in the Synopsis entitled **"SYSTEM FAILURE ANALYZER"** in partial fulfilment of the requirements for the award of the Degree of Bachelor Computer Application/ Bachelor Computer Application (Hons.) with AI & DS of the Graphic Era Hill University, Bhimtal shall be carried out by the undersigned under the supervision of
Dr. HIMANSHU PANT, ASSISTANT PROFESSOR, School of Computing, Graphic Era Hill University, Bhimtal.

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The above mentioned students shall be working under the supervision of the undersigned on the
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Internal Evaluation (By DPRC Committee)

Status of the Synopsis: Accepted / Rejected

Any Comments:

Name of the Committee Members: **Signature with Date**

1.

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ACKNOWLEDGEMENT

I would like to express my heartfelt gratitude to all those who have supported me in the successful completion of this project, **"System Failure Analyzer".**

First and foremost, I extend my sincere thanks to my project guide, **Dr. Himanshu Pant**, for their constant guidance, encouragement, and valuable insights throughout the development of this project.

I also acknowledge the support and motivation provided by the **Department of Computer Applications, Graphic Era Hill University Bhimtal**, which offered the required resources and environment to carry out this project effectively. A special thanks to my group members for their continuous encouragement and feedback during the course of this work.

Lastly, I would like to thank all the authors, researchers, and developers whose works and references helped shape the technical foundation of this project.

This project has been a significant learning experience and a stepping stone in my academic journey.

Synopsis of Project:

Project Title: System Failure Analyzer (SFA)

1. Introduction

1.1 Background

Modern computer systems are required to operate continuously under varying workload. Due to prolonged usage, high resource consumption, or improper maintenance, system may experience failures such as overheating, memory overload, or performance degradation. In many cases, users are unaware of the early warning signs that lead to such failures.

System failures often occur due to a combination of multiple factors rather than a single cause. Identifying these factors manually is difficult, especially for non technical users. Therefore there is a need for a system that can analyze basic system health parameters, access risk, and provide meaningful explanations and preventive suggetions.

A **System Failure Analyze**r is designed to support diagnosis of multiple systems through a centralized analyzer by uniquely identifying each system through a centralized analyzer by ideantifying each system using a unique system ID. It works as a rule based, semi automated system that evaluates system conditions using simulated system metrics, identifies potential failure risks, and suggests corrective actions before critical failure occurs.

1.2 Problem Statement

Many users face unexpected system failures without understanding the underlying cause. Exixsting solutions either require advanced technocal knowledge or provide resultd without clear explanations. Additionally, most basic monitoring tools only display raw data without analyzing or interpreting it.
Therefore, there is a need for an intelligent yet simple system that can:

- Analyze system health parameters,
- Detect possible failure causes,
- Explain the reasoning behing the analysis,
- Suggest preventive measures,
- Maintain Historical record for furture reference

1.3 Need of the Project

- To assist users in understanding system health conditions.
- To detect early warnings signs of system failure.
- To reduce unexpected system crashes through preventive analysis.
- To provide explainable results instead of raw outputs.

- To serve as a foundation for future predictive and intelligence system.

2. Objectives

2.1 Main Objective

The main objective of the **System Failure Analyzer** is to designn and implement a console-based system that analyzes system health parameters, evaluates failure risk, identifies possible causes, and provides preventive recommentations using rule based logic.

- **analyzes system health parameters,**
- **evaluates failure risk,**
- **identifies possible causes,**
- **provides preventive recommentations using rule based logic.**

2.2 Specific Objectives

- To support centralized diagnosis of multiple system using unique system identifiers.
- To collect simulated system metrics such as CPU temperature, CPU usage, RAM usage, Disk usage and system uptime.
- To calculate a risk score based on predefined conditions.
- To classify system status into low, medium, or high risk categories.
- To identify possible system issues automatically.
- To suggest appropriate corrective and preventive actions.
- To store system wise analysis results in log files for historical and comparitive analysis.
- To provide a menu driven and user friendly interface.

3. Literature Review

System monitoring and failure analysis are widely used in operating systems, embedded systems, and enterprise environment. Traditional monitoring tools primarily display real time system statistics, leaving interpretation to the user. Advanced systems use predictive maintenance techniques to reduce downtime and hardware damage.

Research in system relaiability and predictive analytics highlights the importance of historical data analysis highlights and explainable decision making. Rule based systems are often used as the initial stage of intelligent monitoring before integrating machine learning models.

This projects draws inspiration from system diagnostic tools and predictive maintenance concepts but simplifies them for academic implementation using the C programming language.

4. Methodology

4.1 Tools & Technologies Used

- **Programming Language:** C
- Platform: Windows / Linux.
- Interface Type: Console based menu system.
- Data Storage: Text based log files.
- Development Approach: Modular and phase wise.

4.2 Working Flow

1. **Main Menu :**
   - The system displays a console based menu with a multiple options.
   - The user selects a diagnosis option.
2. **System Identification:**
   - The user provides a unique system ID to identify the system being diagnosed.
3. **Metric Collection:**
   - The system collects simulated systeam health parameters such as CPU temperature,CPU usage, RAM usage, disk usage, and uptime
4. **Rule Based Risk Evaluation:**
   - The risk score is initialized and preapared for analysis.
   - Risk points are assigned using rule based logic.
   - Total risk score is calculated.
5. **Risk Classification:**
   - The total risk score is categorized into low, medium, or high risk levels.
6. **Issue Detection and Recommendations:**
   - Possible issues are identified based on abnormal parameter conditions.
   - Preventive and corrective suggetions are generated for the detected issue.
7. **System Report:**
   - The complete diagnosis report is displayed to the user in structured format.
8. **Log Storage:**
   - All analysis details are stored in log files for the future reference and trend analysis

4.3 Block Diagram

User/Central System → System Id + Metrics Input → Rule Based Risk Analysis → Risk Classification → Issue detection & Recommendation Module → Result Display → Log storage (System wise)

5. Project Outcomes

- A functional system failure analysis tool
- Clear identification of system risk levels.
- Explainable diagnosis results.
- Preventive suggetions to reduce failure probability.
- Historical logs enabling trend analysis.
- A scalable base for advanced predictive systems.

6. Application

- Personal computer health analysis.
- Centralized monitoring of multiple systems in labs of offices.
- Diagnostic analysis of PCs, servers and workstations from a single console.
- Educational demonstration of system diagnostics.
- Entry level predictive maintenance system.
- Training tool for understanding system reliability.
- Foundation for embedded and monitoring systems

7. Future Enhancements

- Automatic collection of real system metrics using operating system APIs.
- Remote metric collection using networking.
- Distributed system monitoring.
- Integration of machine learning models for predictive failure analysis.
- Use of Python for data analysis and model training.
- Advanced trend detection and anomaly analysis.
- Graphical user interface and real time alerts.
- Server client architecture for real time diagnosis.
- Deployement as a background service.

8. Conclusion

The **System Failure Analyzer** provides a structured explainable approach to system health analysis. By combining rule based logic, risk assessment, and preventive recommendations, the system helps users understand and manage potential system failures effectively. The project server as a strong academic foundations by applying core programming and analytical concepts as well as reflects industry level practices in system diagnostics and preventive analysis and intelligence systems.

It lays foundation for future enhancements in operating system and machine learning domains.

9. Recources and References

- **C in Depth**, S. K. Srivastava, BPB Publications.
- **GeeksforGeeks** – C Programming Documentation.
- Official C language documentation and tutorials.
- **Operating System Concepts**, Abraham Silberschatz, Peter B. Galvin, Greg Gagne, Wiley.
- Research papers on system reliability and predictive maintenance.
- Online documentation and tutorials related to system monitoring and diagnostics.